RCS01 Student Guide

CTS Cohort 6 | D09 / RCS01 | Monday, July 20, 2026

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big Idea

Residential construction is not just hammering boards together. A building is a set of connected systems. Good tradespeople understand the system, work safely, verify what they did, and leave the work ready for the next person.

Major Residential Systems

System	Plain Meaning	Example
Structure	What holds the building up	Studs, plates, headers, sheathing
Electrical	How power is routed and controlled	Receptacles, switches, lights, breakers
Plumbing	How water enters and waste leaves	Supply lines, drains, traps, shutoffs
HVAC	How air and heat move	Vents, ducting, filters, heat pumps
Envelope	How the building separates inside from outside	Windows, insulation, air sealing
Finish	What protects and completes surfaces	Drywall, trim, paint, flooring

Professional Safety Habits

- Wear PPE that matches the task.
- Ask before using unfamiliar tools.
- Check the work area before starting.
- Use stop-work authority when unsure.
- Tell the truth about mistakes early.
- Clean as you go.

Power Tool Introduction

- Drill/driver: holes, screws, and fasteners. Main hazards: bit slips, stripped fasteners, pinched fingers.
- Circular saw: straight cuts in sheet goods and lumber. Main hazards: blade path, kickback, support, cord path.
- Miter saw: controlled crosscuts. Main hazards: hand placement, blade coast-down, material support.
- Jigsaw/oscillating tool: cutouts and detail cuts. Main hazards: wandering blade, hidden material, hand placement.

Rolling Sink Lab Purpose

Today's lab teaches systems thinking. You will assemble a simple plumbing drain setup, test it, find leaks if they happen, correct them, and document what you verified.

Today's Field Note

A professional is not the person who never makes a mistake. A professional is the person who notices early, stops safely, fixes cleanly, and leaves evidence that the work was checked.

RCS01 Morning Quiz

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Name: _____ Date: 7/20/2026

1. What does RCS focus on in this module?

- A. Only plumbing
- B. Residential construction systems and professional shop habits
- C. Only drywall finishing
- D. Guessing faster

Answer: _____

2. Which item is usually required when using cutting, drilling, or fastening tools?

- A. Safety glasses
- B. Sunglasses only
- C. Loose gloves around rotating tools
- D. No PPE if the task is quick

Answer: _____

3. True or false: Stop-work authority means students can pause a task when something is unsafe or unclear.

Answer: _____

4. Name two major systems found in a residential building.

Answer: _____

5. What should happen before a student uses an unfamiliar power tool?

- A. Watch one internet video
- B. Ask for instructor demonstration and clearance
- C. Try it quickly
- D. Let the fastest student go first

Answer: _____

6. What is the purpose of a P-trap?

- A. It stores clean water for drinking

- B. It holds water to block sewer gas and creates a service point
- C. It increases water pressure
- D. It replaces the faucet

Answer: _____

7. Why should parts be dry-fit before final tightening?

- A. To make the lab take longer
- B. To check alignment and avoid forcing parts
- C. To skip inspection
- D. To hide leaks

Answer: _____

8. Which is a stop condition during the sink lab?

- A. A small leak appears
- B. A washer is missing
- C. The group cannot explain the path
- D. All of the above

Answer: _____

9. What is tool accountability?

Answer: _____

10. Complete the sentence: A professional does not guess. A professional _____.

Answer: _____

RCS01 Afternoon Lab - Rolling Sink Module

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Lab Mission

Build a simple sink drain and P-trap assembly in the rolling lab module, inspect it, leak-test it, troubleshoot problems, and document verified work.

Safety Boundary

- Water stays off until the assembly has been inspected.
- No electrical tools or cords near wet areas.
- Wipe spills immediately.
- Stop if a fitting does not align or a part is missing.
- Do not overtighten plastic fittings. Hand-tight plus instructor guidance only.

Roles

Role	Job
Builder	Assembles parts and calls out each step.
Inspector	Checks alignment, washers, tightness, and leaks.
Documenter	Records parts, issues, fixes, and evidence photos.

Parts Inventory

Part	Have it?	What it does	Notes
Sink basin/module		Holds water and routes it to drain	
Faucet		Controls water into sink	
Supply line/shutoff		Brings water to faucet and allows shutoff	
Tailpiece		Connects sink drain to trap	
P-trap		Holds water seal and service point	
Slip nuts/washers		Seal adjustable drain joints	
Drain outlet		Carries water away	

Build Sequence

1. Inventory all parts.

2. Lay parts out in system order.
3. Dry-fit the tailpiece, trap, and outlet path.
4. Check washer direction and slip nut alignment.
5. Hand-tighten fittings.
6. Stop for inspection before water testing.
7. Place bucket/towel before testing.
8. Run a small amount of water and observe.
9. Mark or record any leak before fixing it.
10. Correct one issue at a time.
11. Retest and verify dry joints.
12. Reset the module after instructor approval.

Leak Troubleshooting Table

Problem observed	Likely cause	Fix attempted	Retest result	Instructor initials

Success Criteria

- Parts identified and assembled in correct order.
- No leaks after correction and retest.
- Group can explain the water path.
- Station is clean, dry, and reset.
- Verification log is complete.

RCS01 Verification Log

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Student: _____ Group: _____ Date: 7/20/2026

Task / Station	What I expected	What I observed	Method used	What I verified	What still needs checking	Signoff
PPE check						
Tool / part inventory						
Sink assembly inspection						
Leak test						
Cleanup/reset						

Sample Entry

Task: Sink assembly inspection | Expected: washers seated and slip nuts straight | Observed: one washer missing | Method: visual inspection before water test | Verified: corrected washer installed before testing | Still needs checking: leak test after water is on

RCS01 Exit Ticket

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Name: _____ Date: 7/20/2026

One residential system I can name is:

_____

_____

One safety rule I need to remember is:

_____

_____

One thing I verified during lab was:

_____

_____

One thing I want more practice with is:

_____

_____